GENERAL SPECIFICATION G4.38

Fine Bubble Air Diffuser System

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G4.38 FINE BUBBLE AIR DIFFUSER SYSTEM

This specification covers the general requirement of the performance, design and construction of the Fine Bubble Air Diffuser system and its associated plant and equipment in used water (wastewater) application. The system shall comply with all relevant statutory regulations and the latest edition of all relevant international, harmonised European and British Standard and Singapore standard and code of practices.

This specification shall also comply with:

- The relevant particular specification;
- The relevant Electrical specification; and
- The relevant Instrumentation Control and Automation (ICA) specification.

G4.38.1 Definitions

(a) Adequate Mixing: Variation in mixed liquor suspended solids (total residue) of less than 10 percent between the mean value of samples taken at any two depths along any vertical line extending between water surface and elevation of the top of diffusers.

(b) Aeration Tank: Structure with several anoxic and aeration zones.

(c) Zone: Portion of the Aeration Tank separated from other portions of the aeration tank for process design and operation. A zone may contain one or more aeration grids.

(d) Diffuser Assembly: Membrane disc diffuser with an element holder and retaining device.

(e) Distribution Header: Piping between manifold and diffuser assembly.

(f) Dropleg: Connection from air source to manifold.

(g) Dynamic Wet Pressure (DWP): Pressure to operate at specified conditions minus submergence and flow control losses.

(h) Grid: Configuration of diffuser systems within basin.

(i) Manifold: Single run of piping that connects dropleg or cooling header with distribution header(s).

(j) Normal Cubic Metres per Hour (Nm^3/h): Air flowrate standardised to 20 degrees C, 101.35 kPa, and 0 % relative humidity, unless otherwise specified.

(k) Standard Oxygen Transfer Rate (SOTR or SOR): Rate of oxygen transfer to tap water at standard conditions of 20 degrees C, 0.0 mg/l residual dissolved oxygen concentration, and a barometric pressure of 101.35 kPa (dry air).

G4.38.2 Submittals

(a) The Contractor shall provide complete fabrication and assembly drawings, together with detailed specifications and data covering materials, power drive, assemblies, and accessories forming a part of the equipment supplied. The data and specifications for each unit shall include, but shall not be limited to, the following:

(i) Outline and general arrangement drawing of the equipment indicating the designation overall dimension / footprint and typical layout of key components / system;

(ii) Overall dimensions and estimated weights of all furnished equipment;

(iii) Component drawings showing all details of construction;

(iv) A typical control philosophy;

(v) A typical commissioning plan (Including timescale for completion);

(vi) Detailed mechanical drawings showing equipment fabrications and interface with other items. Include dimensions, size, and locations of connections to other work, and weights of associated equipment;

(vii) Manufacturer's catalogue information, descriptive literature, specifications, and identification of materials of construction;

(viii) A detailed drawing of proposed aeration equipment layout for each Basin showing airline sizes and lengths, distances between air distribution headers, and location of diffusers, supports, expansion joints, purge lines / valves, instrumentation, etc;

(ix) Diffuser, diffuser connector, balancing orifices, and system head loss curves covering range of airflow rates specified;

(x) Calculations showing distribution and balancing of air within each Basin for minimum and maximum airflow rates specified;

(xi) Calculations by a registered Professional Engineer (P.E.) to demonstrate that support design complies with requirements of this Section;

(xii) Shop and Field Painting Systems Proposed: Include manufacturer's descriptive technical catalogue literature and specifications;

(xiii) Identification, size, rating and location of all instrument and electrical connections;

(xiv) The Energy consumption and electrical drive loadings list;

(xv) Wiring diagrams which show internal wiring, external connection wiring and interconnection wiring requirements. Manufacturer shall indicate the power supply and control power supply requirements;

(xvi) Specification for diffusers, piping and supports including materials, ratings, etc;

(xvii) Details of lifting / removal systems where required. This includes supports, guides, chains and fixing points, lifting equipment requirements and detailed methodology for removal and replacement of aeration grids;

(xviii) List of all instruments and controls, along with function; and

(xix) Make, model, and weight of each equipment assembly.

G4.38.3 Manufacturer's Qualifications

(a) The Contractor shall provide a reference list for the specific proposed diffused air system (manufacturer, type, model), as per qualifications stated in Performance Specification.

(b) The manufacturer shall submit an installation list with contact information for the equipment quoted verifying to the S.O. that the specified manufacturer's qualifications are met.

(c) Manufacturer shall submit the Manufacture's Certificate of conformance for manufactured commercial products.

G4.38.4 Design Specification- General

(a) Manufacturer shall furnish the complete header, fine bubble, and diffused aeration equipment system as a complete package. Including, but not necessarily limited to droplegs, air manifolds, distribution headers, diffuser elements, pressure monitoring systems, supports, drain lines, purging systems, header joints, expansion joints, and accessories and connections for cleaning system.

(b) The Contractor shall ensure that the complete diffused air system is designed and supplied by a single specialist manufacturer with qualifications as noted above, and that the specialist manufacturer has full responsibility for the system functionality and compliance with this Specification. All fine bubble diffused air systems shall be supplied by this single manufacturer.

(c) Furnish complete, engineered systems. Drawings indicating the locations of droplegs, air manifolds, distribution headers and other associated equipment. The diffuser manufacturer is to identify locations of features such as floor slopes, baffles, un-aerated areas required for the Mixers etc., and co-ordinate the design so as to avoid installation conflicts. Details of droplegs, air manifolds, distribution headers, header spacing, diffuser spacing, supports, and appurtenances shall be defined by and be the responsibility of Contractor and shall be consistent with the requirements in this Specification.

(d) Droplegs support system shall allow for expansion and contraction over a temperature range of 60 to 135 degrees C when installed.

(e) Manifold Header and Header Support System shall allow for expansion and contraction over a temperature range of 60 to 135 degrees C (mean wall temperature) when installed.

(f) Expansion/contraction system consisting of fixed or flanged joints and guide supports shall be included. Guide supports shall allow for longitudinal movements. The expansion bellows material shall be type 316 stainless steel unless otherwise specified.

(g) Gaskets shall be EPDM, 45 to 55 durometer as minimum; locate at expansion joints and couplings to form an airtight connection.

(h) Nuts, bolts, washers, and other non-welded parts shall be Type 316 stainless steel, ASTM A240. Threaded assemblies shall be chemically treated or lubricated prior to assembling to prevent galling. Alternative materials such as glass-filled polypropylene (GFPP) guaranteed to provide the required strength with allowance margin and guaranteed against corrosion shall also be acceptable on the condition of significant reduction of installation duration. The enhanced ease and reduced installation reduction shall be justified to the S.O..

(i) Anchor Bolts shall be Type 316 stainless steel, for metal fabrications and casting shall be use epoxy placed anchor bolts. Wedge anchors are not acceptable.

(j) Unless otherwise specified, insulating Flanges shall be galvanically compatible with piping, materials shall be resistant for the intended exposure, operating temperatures, and products in the pipeline. Gaskets shall be full-face Type E with O-ring seal. The flanged gasket shall be supplemented with a neoprene facing on each side to accomplish a seal. Insulating Sleeves and washer shall be full-length fibreglass reinforced epoxy.

G4.38.5 Aeration Equipment

(a) Diffuser Manufacturer Interface for Aeration Equipment

(i) Manufacturer shall supply complete aeration equipment for each grid below the aeration tank coping level. Equipment to be supplied by the diffuser manufacturer above the coping shall be extensions of equipment necessary to provide a complete aeration system below the coping. This will include equipment such as diffuser pressure monitoring systems and airlift purge systems.

(b) Aeration Grids

(i) Each grid shall be designed to be geometrically similar with respect to manifolds, distribution headers, diffuser holder elements, drain line, sump, airlift purge, and pressure monitoring systems.

(ii) Minimum one (1) pressure monitoring system shall be provided for each aeration grid.

(iii) Condensate / water purge lines shall be provided for each aeration grid. The purge lines shall allow operators to be able to visually inspect the colour of the purge fluid to verify the integrity of the diffusers and distribution piping.

(iv) Aeration grid for mixed liquor channel shall be removable, even while the channel is full of mixed liquor and operational. Grid piping shall be in sections and connected by quick coupling dismantling joints. Lifting davit shall be provided for removing of the pipes.

(c) Droplegs

(i) 316SS droplegs shall be provided, terminating in flanged connections above the tank coping level. The droplegs shall be designed to ensure adequate cooling of air is provided before entering the aeration distribution piping and diffusers, such that temperatures do not exceed manufacturer recommendations and that diffuser life guarantees are in no way compromised.

(ii) Sufficient and adequate expansion joints and insulating flanges shall be provided to prevent thermal movement exerting stress on fixed supports and structures.

(d) Air Manifolds

(i) The air is supplied through the main air piping to each aerated zone. Each zone shall be equipped with a flow control valve and each dropper shall be equipped with a manual isolation valve. The flow control valve shall be selected and sized to provide accurate flow control over the air flowrate range required to achieve the specified SOTR range. It shall not be simply a line size valve.

(ii) Air manifold shall be located at end of distribution headers only.

(iii) Connections shall be provided to enable effective in-situ cleaning of each aeration grid and its associated diffusers with the zone remaining full of mixed liquor.

(iv) Provide a minimum of one drain line and airlift purge system for air manifold and distribution headers. The purge systems shall include collection sumps within the manifold and distribution piping to ensure the piping can be fully drained and purged of liquid while the diffuser system is operational.

(v) Manifolds shall be provided for long-term exposure to 60 degrees C mean-wall temperature.

(vi) Expansion/contraction system consisting of fixed or flanged joints and guide supports shall be included. Guide supports shall allow for longitudinal movements.

(e) Distribution Pipe and Distribution Headers

(i) Distribution headers with identical diffuser holder spacing shall be provided for all grids located in the aeration tanks.

(ii) Orient distribution headers perpendicular to air manifolds and parallel to the direction of process flow.

(iii) All distribution headers and fully submerged distribution pipe shall be Schedule 80 cPVC, unless otherwise specified.

(iv) Fabricate in sections up to a maximum length to suit the specified area in the drawings, with fixed joints, non-rotational joints, or flanged joints.

(v) Fabricate with diffuser element holders, factory solvent-welded to crown of header. Attach diffuser elements to distribution headers to resist 200 N m applied torque about polar axis of holder and 135 Nm about longitudinal axis.

(f) Diffuser Assemblies

(i) Membrane disc diffuser assemblies shall be provided including diffuser elements, O-rings, retainers, PVC holders, factory solvent welded to distribution headers. Additional elements shall be provided as required to meet the specified performance.

(ii) Alternatively, diffuser holder shall be fitted on a modular saddle mounted onto the piping system.

(iii) Diffuser assemblies in diffuser holders shall be installed as specified to meet the system and aeration tank performance requirements for each grid. Where diffuser assemblies are not provided, orifice plug for the diffuser holders shall be provided.

(iv) Diffuser elements not installed shall be packaged for long term storage.

(g) Membrane Disc Diffuser Assemblies

(i) Circular Membrane Disc Diffuser Elements

1) Membrane disc diffuser elements shall meet the tensile strength, elongation and hardness to ensure reliability and performance during operational lifespan;

2) The diffusers shall be free of tears, voids, bubbles, creases, or other structural defects;

3) Free of loose or unbonded material, cracks, chips, spalling, or other structural defects. Diffuser elements shall be free of any material soluble in wastewater;

4) Uniform distribution of air bubble release across active surface (horizontal projected area) of diffuser element when submerged in water;

5) Membrane diffuser shall have integral check valve function to prevent backflow of wastewater into the aeration grid;

6) Dynamic Wet Pressure (DWP) systems shall be proposed to S.O. for approval; and

7) Sufficient strength to support a hydrostatic load equivalent to the specified water depth with a safety factor of two.

(h) PVC Diffuser Element Holders shall comply with the following:

(i) Air plenum chamber below diffuser element;

(ii) Mechanism to attach diffuser element to element holder; and

(iii) Provide complete peripheral edge support for membrane disc diffuser element.

(i) Retaining Devices shall comply with the following:

(i) Securely hold and seal membrane disc diffuser element to element holder;

(ii) Diffuser assembly and retaining device shall prevent air escape at diffuser element-sealing interface;

(iii) Vertical edges of diffuser elements shall not be exposed to liquid;

(iv) Sealing of membrane diffuser shall provide a long-term positive seal and prevent air escape except through active area of diffuser element;

(v) Force applied to the diffuser membrane shall be able to be monitored / optimised during installation; and

(vi) Each diffuser element holder shall have an airflow control orifice.

G4.38.6 Pipe Supports

(a) All support and wetted parts shall be fabricated from type 316 stainless steel, unless otherwise specified.

(b) Header vertical adjustment shall be continuous and possible without removing air piping from support. The header adjustment shall be within ±15 mm lateral and ±52 mm vertical.

(c) Each air piping section shall have a minimum of two supports and additional supports as necessary to maintain level. Support height shall be sufficient to provide diffuser elevations as specified. Each support shall provide a bearing surface contoured to fit 360 degrees of air piping. Bearing surface shall be a minimum of 52 mm wide for manifolds and 40 mm wide for distribution headers.

(d) Dropleg Supports

(i) Sliding Supports

1) Locate sliding supports directly adjacent to expansion joints. Locate sliding supports at horizontal ends of droplegs. Maximum spacing between sliding supports shall be 2.5 m in vertical and horizontal runs of piping;

2) Sliding support shall allow for longitudinal movement of pipe due to thermal expansion. Supports shall allow a 3 mm clearance around dropleg to permit pipe movement;

3) Design sliding supports to resist uplift due to buoyant forces with a minimum safety factor of 5.0; and

4) Attach sliding supports to walls and floors with appropriately sized anchor bolts consistent with other sections of this specification.

(ii) Anchor Supports

1) Locate one anchor support for each dropleg as described herein;

2) Anchor support shall be adequate to resist longitudinal and lateral pipe movement due to thermal expansion at the stated differential temperature described herein; and

3) Anchor dropleg to wall with appropriately sized anchor bolts consistent with other sections of this Specification.

(e) Air Manifold Piping Supports

(i) Maximum spacing between supports of 2.5 m;

(ii) Resist thrust generated by expansion or contraction of air distribution headers; and

(iii) Include manifold hold-down, guide straps, anchor bolts and supporting structure. Guide straps shall resist the uplift force per support without exceeding the design stress.

(f) Air Distribution Header (Guide) Supports

(i) Maximum spacing between supports of 2.25 m;

(ii) Allow longitudinal movement of header section to prevent stress build-up in header due to thermal expansion/contraction forces;

(iii) Consist of self-limiting hold down and sliding mechanism. Sliding mechanism shall provide minimum resistance to movement of air distribution header under full buoyant uplift load. Mechanism shall provide 3 mm clearances around header and be self-limiting if mechanism is overtightened. Design shall consider the maximum horizontal thrust that will initiate movement of header relative to mechanism under full buoyant uplift load; and

(iv) All support shall be design and endorsed by P.E and submitted to S.O. for approval. All authorities' submission shall be contractor responsibility.

G4.38.7 Drain line, Sump and Airlift Purge

(a) Purging / Draining Systems

(i) The system shall be able to drain entire submerged aeration piping system of all water while in normal operation. Each grid shall be equipped with one cPVC drain line, sump, airlift purge eductor line and eductor carrier column. Sump location shall be at the invert of the manifold piping.

(b) Sumps

(i) Bottom elevation shall be lower than invert of air distribution headers, and drain line to ensure complete drainage is achieved; and

(ii) Connect each sump to an airlift eductor line, same material as drain line, extending to above the aeration zone top water level (so that the purging fluid can be observed by operators).

G4.38.8 Diffuser Pressure Monitoring System

(a) Each aeration grid shall be provided with one pressure monitoring connecting box, bubbler tube, support brackets, polyethylene tubing and carrier pipe to enable monitoring of dynamic wet pressure using a portable device.

(i) Connecting Box shall be independent, stanchion-mounted fibreglass box capable of containing three 12-mm PVC ball valves, and quick coupling connectors, all of which are accessible through a front access door;

(ii) All Mounting Hardware shall be Type 316 stainless steel;

(iii) Mount pressure monitoring systems above top of aeration tank coping. Stainless steel floor sleeves and mechanical link seals shall be provided. The exact location of the diffuser to be determined by Contractor / Manufacturer; and

(iv) One portable monitoring panel shall be provided with at least one differential pressure gauge, quick coupling connectors, PVC ball valves, one set of calibration curves, and tubing and fittings as necessary to measure pressure differential between header and holder plenum and holder plenum and diffuser submergence.

G4.38.9 Header and Manifold Joints

(a) Provide positive type connection joints bolted or flanged or threaded union type for all submerged header and manifold joints.

(b) Bell and spigot, slip-on or expansion joints shall not be utilised for submerged joints. All joints must be of the positive locking type.

(c) All fixed joints shall have interlocking spline teeth that engage with matching protrusions such that rotation of air distributors is prevented. All rotational forces shall be transferred through the interlocking splines. The splines shall also allow for air distributor axial orientation adjustment during installation. An O-ring gasket and a threaded screwed-on retaining ring shall be used to prevent air leakages and further secure the connection. Joints that make use of the O-ring between the joint interfaces to prevent air distributor rotation shall not be accepted.

(d) Threaded union joints shall consist of a spigot section solvent welded to one end of a distribution header, a threaded socket section solvent welded to the mating distribution header, an “O” ring gasket and a threaded screw on retainer ring. Solvent welding shall be done in the factory. Flanged joints shall be of the follower-type with stainless steel hardware and shall have standard drilling.

G4.38.10 Fabrication and Corrosion Protection

(a) Shop fabricate welded metal parts and assemblies from type 316 stainless steel, unless otherwise specified.

(b) Clean all stainless-steel surfaces after fabrication and welding by using the following procedure:

(i) Pre-clean all outside weld areas to remove weld splatter with the use of stainless steel brushes and/or deburring and finishing grinding wheels;

(ii) Finish clean to remove all carbon deposits, light oxide films and similar contaminants by full immersion pickling and rinsing, to aid the regeneration of a uniform corrosion resistant chromium oxide film as follows:

1) Completely immerse all stainless steel assemblies and components in an acid solution as described in Section 6.2.11 of ASTM A380-88. The acid shall be a nitric-hydrofluoric solution as defined in Table A.2.1 of Annex A2 of ASTM A380; and

2) Provide a final thorough rinse using ordinary industrial or potable water and dry in conformance per Section 8.3 of ASTM A380.

(iii) Corrosion protection techniques not using full immersion methods will not be acceptable; and

(iv) At least two weeks prior to cleaning equipment, the Contractor shall notify the S.O. of the schedule so that the S.O. can observe the procedure if desired. If full immersion pickling is not observed by the S.O., the Manufacturer shall verify that cleaning has been performed per the above specification. Non-properly cleaned equipment will be cause for rejection of the system.

G4.38.11 Performance Requirements

In additional to the specific performance requirement stated in particular specification, the following shall be required:

(a) Airflow Rate Output

(i) Not differ by more than 10 percent, at minimum and maximum airflows, for any two diffusers in a single aeration zone (based on diffuser with lower flow rate).

(b) Air Distribution and Balancing shall be controlled by use of orifices and proper header size selection only.

(c) Adequate mixing throughout the aeration tanks at the stated minimum airflow requirements shall be achieved.

(d) Adequate mixing throughout the aeration tanks at mixed liquor suspended solids concentrations up to 10,000 mg/L.

(e) Air Distribution and Balancing

(i) It shall be sufficient to maintain mixed liquor suspended solids in a state of suspension over entire depth of aeration Basin at stated minimum airflow requirements.

(f) Individual Aeration Tank shall meet the following additional Performance Requirements

(i) Provide complete calculations showing how the system will operate to meet the Peak and Average Design SOTRs shown in the Particular Specifications. As a minimum, provide calculations showing SOTE, airflow rate per diffuser, grid airflow rates, and pressures at the top of each dropleg as well as allowance for pressure increase due to diffuser fouling; and

(ii) Provide recommendation of the minimum airflow rate per diffuser to avoid entrainment of water into the diffuser and cause subsequent diffuser fouling. Provide complete calculations showing how the system will operate at the minimum airflow per diffuser and identify any grids that may be mixing limited. As a minimum, provide calculations showing SOTE, airflow rate per diffuser, grid airflow rates, and pressures at the top of each dropleg.

G4.38.12 Factory and Performance Testing

(a) Factory Materials Testing

(i) The following Quality Control test shall be performed at the point of manufacturer, prior to membrane perforation:

1) Durometer Hardness Test, Shore A, per ASTM 2240;

2) Tensile Strength Test, per ASTM D412; and

3) Minimum Modulus of Elasticity Test, per ASTM D412.

(ii) The following Quality Control test shall be performed at the point of manufacture, after membrane perforation:

1) Dynamic Wet Pressure Test; and

2) Air Flow Uniformity Test.

(iii) Any other test deemed necessary to prove the integrity of the membrane.

(b) Field Functional Test

(i) Conduct bubble testing on each aeration grid at water level nom 300 to 500 mm above the diffusers. This test shall demonstrate even distribution of fine bubbles across each aeration zone and complete absence of large bubbles that may indicate leaks in piping, diffuser defects, incorrect diffuser membrane seating, etc;

(ii) Conduct flow, pressure and Mixing tests on each aeration grid of the aeration tanks;

(iii) Perform under actual or approved simulated operating conditions. Airflow shall be as measured by treatment plant instrumentation. Calibrate airflow instrumentation as part of testing procedure;

(iv) Test for a continuous 3-hour period without malfunction;

(v) Adjust, realign, or modify units and retest if necessary;

(vi) Pressure Test: Measure air pressure in dropleg above top of aeration tank coping, and at maximum airflows and submergences as per Performance Requirements in particular specification.;

(vii) Air temperature testing: Measure air temperature in aeration grid pipework to confirm sufficient air cooling is provided by the dropleg piping or other systems. This may require installation of permanent temperature measurement primary devices; and

(viii) Mixing Test

1) Perform at minimum recommended airflows for each grid as per Performance Requirements in particular specification;

2) Select three vertical lines and two depths in each basin;

3) Take three Samples at each of two depths along each vertical line;

4) Independent testing laboratory approved by S.O. will perform residue test on each Sample. Mean value of total residue for three Samples at each depth will be used to determine conformance with requirements;

5) All testing and sampling procedures shall be submitted to S.O. for approval prior the test commence; and

6) Contractor to provide safe access for all field functional testing.

(c) Factory Oxygen Transfer Performance Test

(i) A mass transfer test at the manufacturer’s test facility or at an independent test facility shall be required to verify the aeration equipment’s clean water Standard Oxygen Transfer Rate (SOTR) at the maximum condition stated in Particular Specifications. This testing shall for the specific diffuser density / pattern and depth for each of the different aeration zone designs.

(ii) A graph will be prepared from this data to be used in determining the expected performance of the diffusers at other flow conditions. This graph will consist of SOTE per meter of submergence depth plotted versus Nm^3/hr per diffuser. The Contractor shall notify S.O. at least 30 days prior to conducting the oxygen transfer performance test.

(iii) The Manufacturer shall bear the cost of all expenses required to satisfactorily complete the aeration equipment oxygen transfer performance tests, including but not limited to labour, testing instruments, equipment, power, and materials.

(iv) The aeration equipment's SOTR shall equal or exceed the SOTRs required to meet the maximum design condition specified in Particular Specifications.

(v) All mass transfer tests used as a basis for verification of the aeration equipment performance shall be done by non-steady-state re-aeration. A mass transfer test shall consist of three re-aeration test runs. The SOTR for each flow condition shall be the average of the SOTRs obtained for each re-aeration test run. Sodium sulphite by cobalt shall be used to strip residual dissolved oxygen between re-aeration test runs.

(vi) All performance testing shall be conducted in test facilities provided by the manufacturer, or as approved. The test facility shall be capable of providing the side water depth and diffuser submergence specified hereinbefore. The diffuser density (diffusers per square metre of tank area) shall be equal to the diffuser density required for each aeration grid tested.

(vii) The testing procedures to be used in determining the oxygen transfer capacity of the aeration equipment shall be as described in the ASCE Standard for the Measurement of Oxygen Transfer in Clean Water. Specific details of the test procedure shall be submitted for review and approval of the S.O..

(viii) The testing procedure may include a 5% variance on the air rate required to achieve the specified SOTRs.

(d) In-Situ SOTR Testing

(i) An in-situ SOTR test shall be conducted, if specified, and shall comply to the following:

1) To demonstrate the maximum SOTR is achievable at the contractor specified / guaranteed air flowrate;

2) The test shall be carried out in accordance with ASCE/EWRI 2-06; and

3) The contractor shall be responsible for supply of all chemicals, labour and instrumentation as well as analysis of testing results. The Principal will be responsible for filling the zone to the required level with potable water.

(e) Performance Testing Results

(i) The Contractor hereby accepts the test procedures specified in this Section as a valid basis for determining the ability of the aeration equipment to meet the Performance Requirements specified herein;

(ii) Obtain approval of test reports from S.O. prior to shipment of any equipment;

(iii) In the event the aeration equipment fails to meet the dissolved oxygen Performance Requirements specified, the Board shall have the right to require the Contractor to modify or replace the aeration equipment to enable said system to meet the Performance Requirements. Field testing of the modified or replaced aeration equipment shall be performed to ensure compliance with the dissolved oxygen requirements;

(iv) The first field test, if required, shall be performed at the Contractor's sole expense;

(v) All costs arising from project delays caused by the failure to meet the Performance Requirements specified herein shall be borne by the Contractor;

(vi) A report documenting the procedures used and results of all performance tests, including factory and field tests, if required, shall include all initial measurements, computations, and final results, shall be submitted in accordance with Article Submittals herein; and

(vii) Any publication or release of the performance testing data referencing the Board shall be done only with prior approval of the Board.

G4.38.13 Training

As a minimum, provide the following training:

(a) Two sessions of classroom training for the Board's operation personnel;

(b) Two sessions of classroom training for the Board's mechanical maintenance personnel;

(c) Two sessions of classroom training for the Board's electrical maintenance personnel;

(d) Two sessions of hands-on training for operation personnel; and

(e) Two sessions of hands-on training for the Board's maintenance personnel.

G4.38.14 Special Installation Requirement

Special installation requirements include the following:

(a) Appropriately skilled representatives of the diffuser system manufacturer shall be present during system installation to confirm all design requirements are met, the installed system is adjusted to within the correct tolerances and no damage is caused to diffuser membranes or other systems. The diffuser system manufacturer shall provide written approval of the system installation and that the installation is complaint with the requirements for all performance, life and other guarantees.

(b) Piping and diffusers shall be adequately protected from exposure to UV light and heat prior to the filling of the tanks with mixed liquor.

(c) Where removable diffuser grids are required, demonstrate removal and replacement of each grid with the tankage at normal operating level.

END OF SECTION